

# *Deep Learning: Theory and Practice*

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## Deep Learning - Practical Considerations

02-04-2020

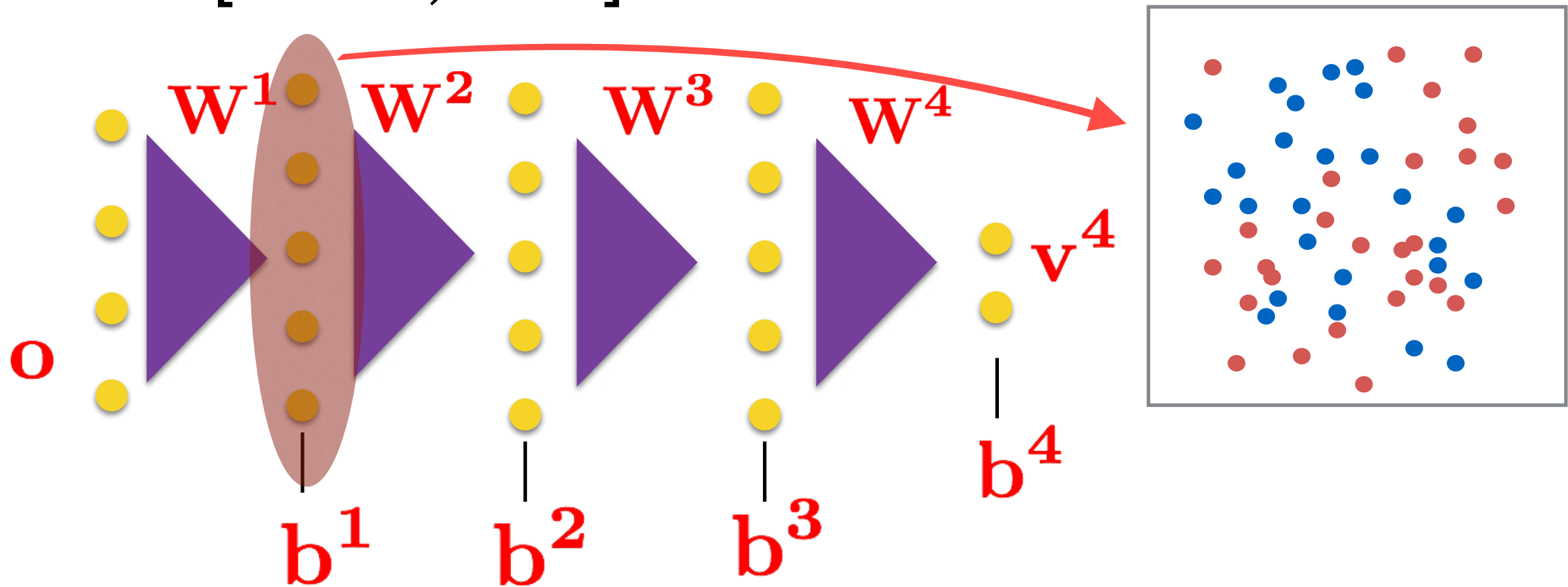
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*deeplearning.cce2020@gmail.com*



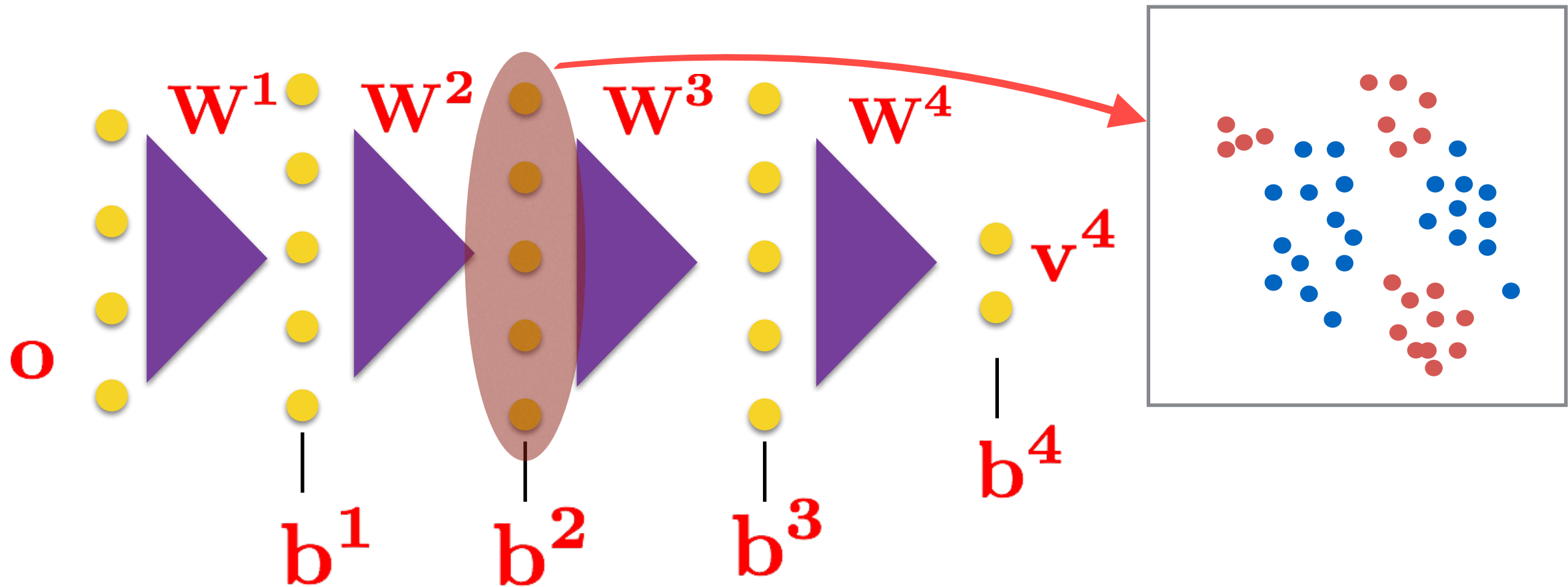
# Deep Networks Intuition

Neural networks with multiple hidden layers - Deep networks [Hinton, 2006]



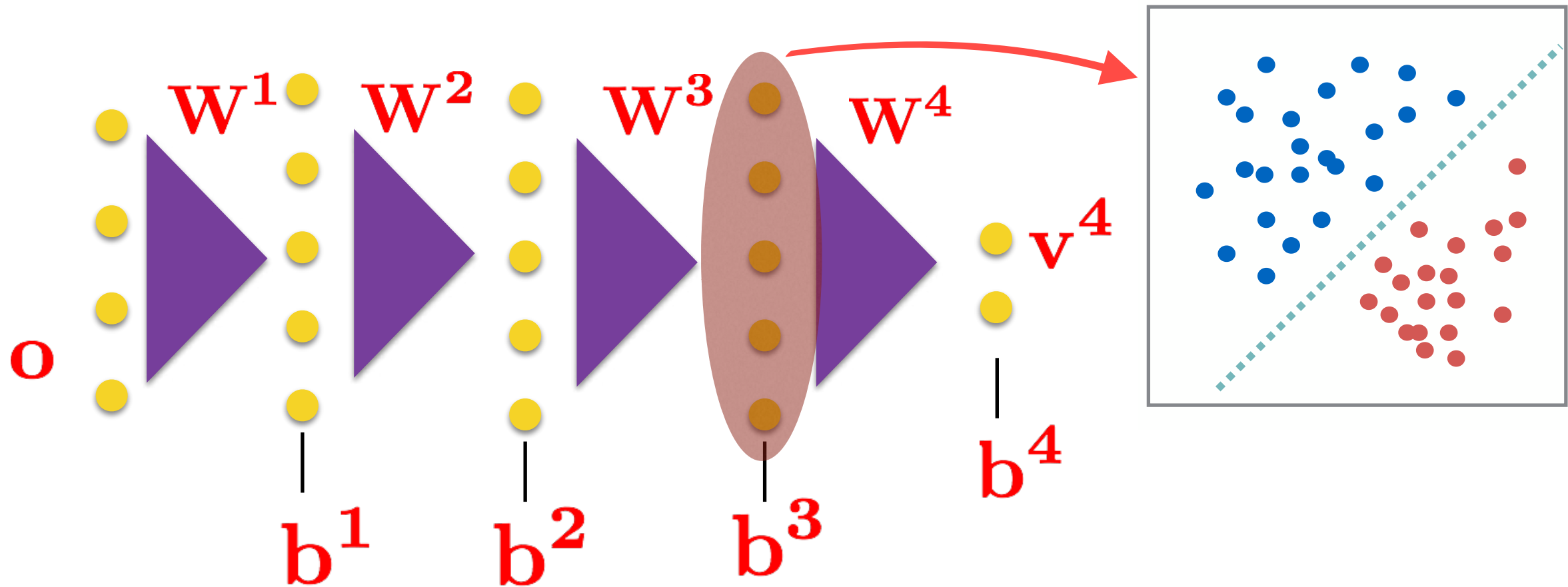
# Deep Networks Intuition

Neural networks with multiple hidden layers - Deep networks



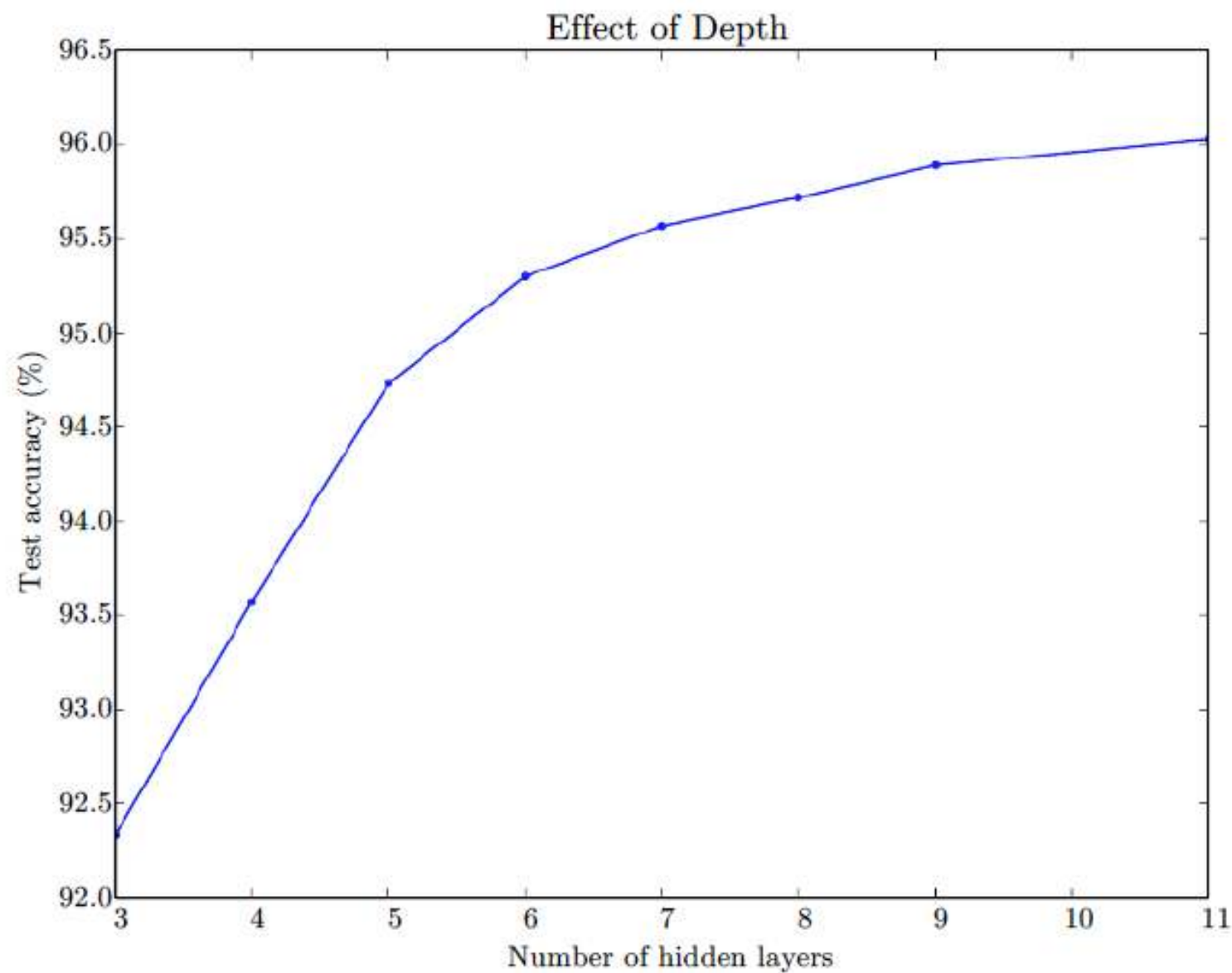
# Deep Networks Intuition

Neural networks with multiple hidden layers - Deep networks



Deep networks perform **hierarchical data abstractions** which enable the non-linear separation of complex data samples.

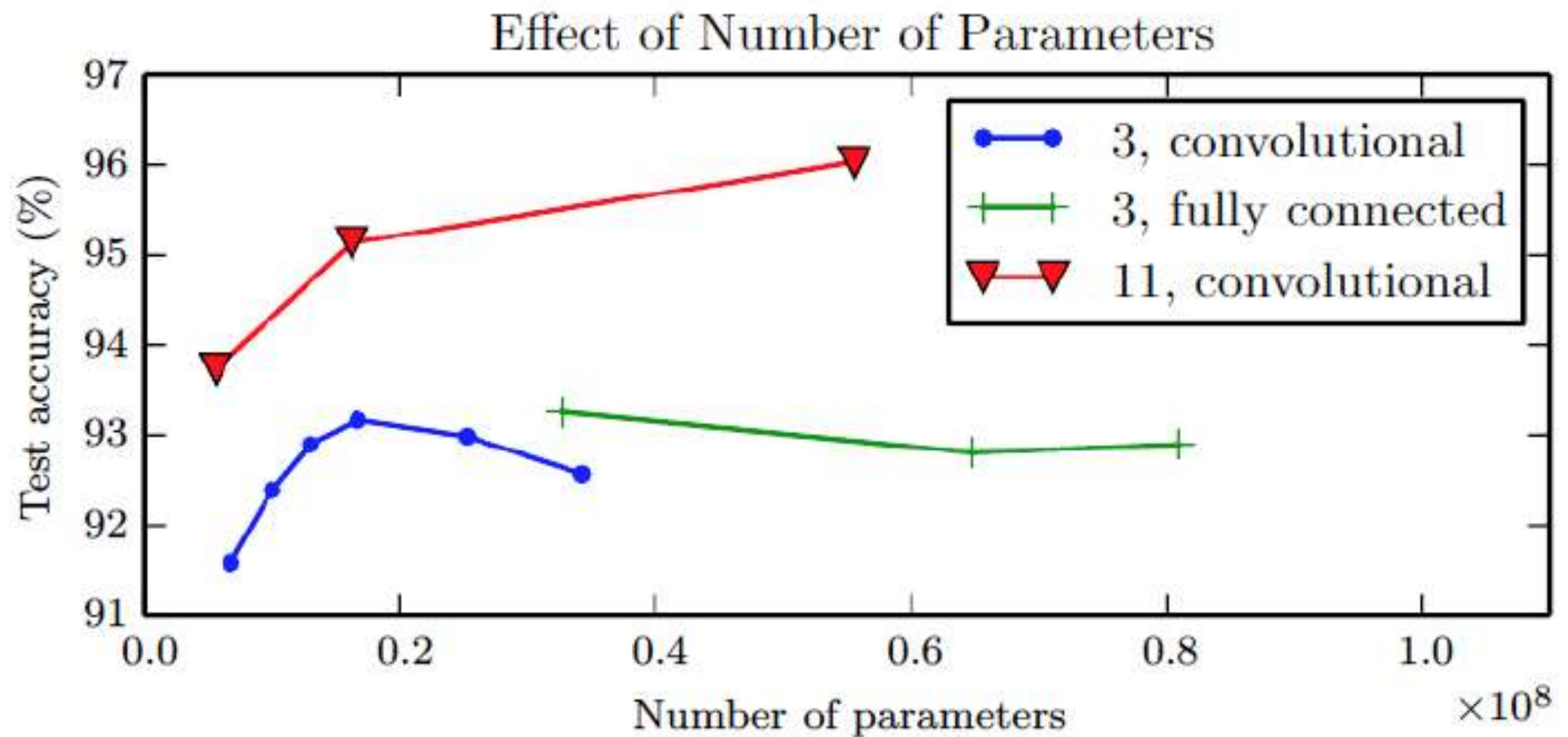
# Need for Depth



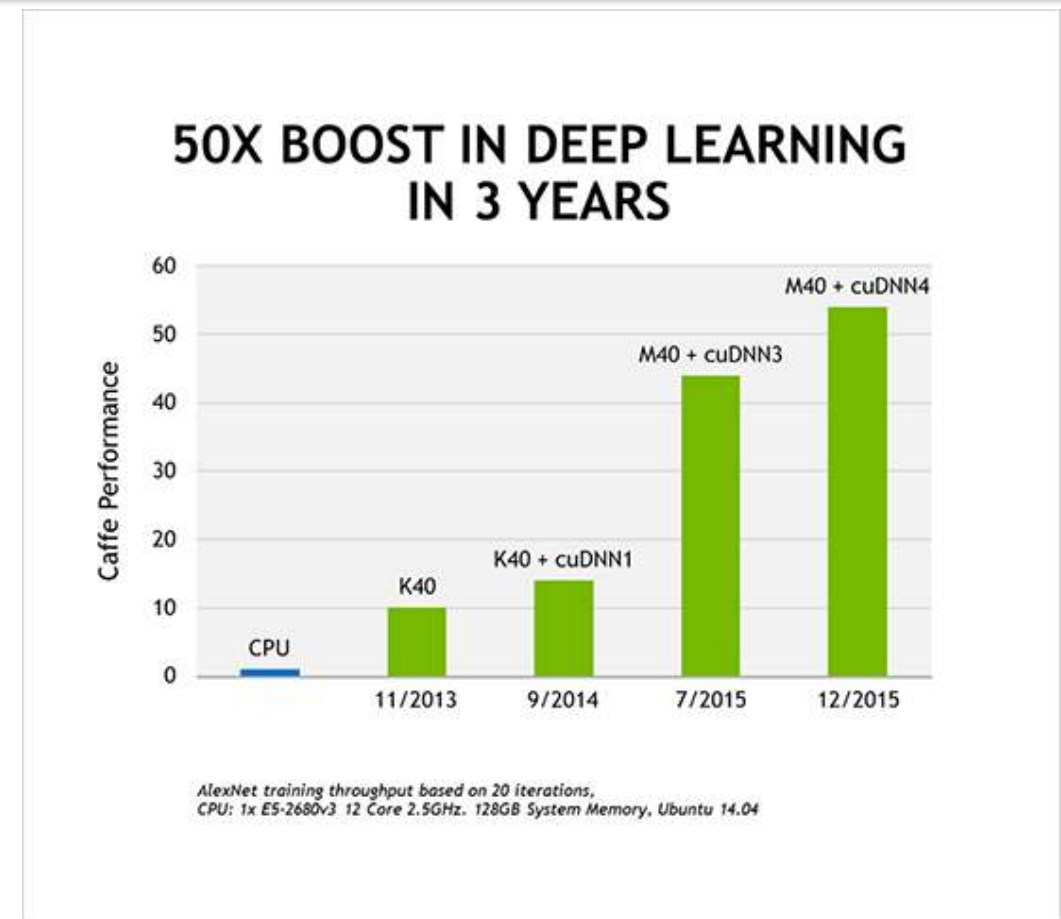
$$\mathbf{h}^{(1)} = g^{(1)} \left( \mathbf{W}^{(1)\top} \mathbf{x} + \mathbf{b}^{(1)} \right)$$

$$\mathbf{h}^{(2)} = g^{(2)} \left( \mathbf{W}^{(2)\top} \mathbf{h}^{(1)} + \mathbf{b}^{(2)} \right)$$

# Need for Depth



# Deep Networks



- Are these networks trainable ?
  - Advances in computation and processing
  - **Graphical processing units** (GPUs) performing multiple parallel multiply accumulate operations.
  - Large amounts of supervised data sets

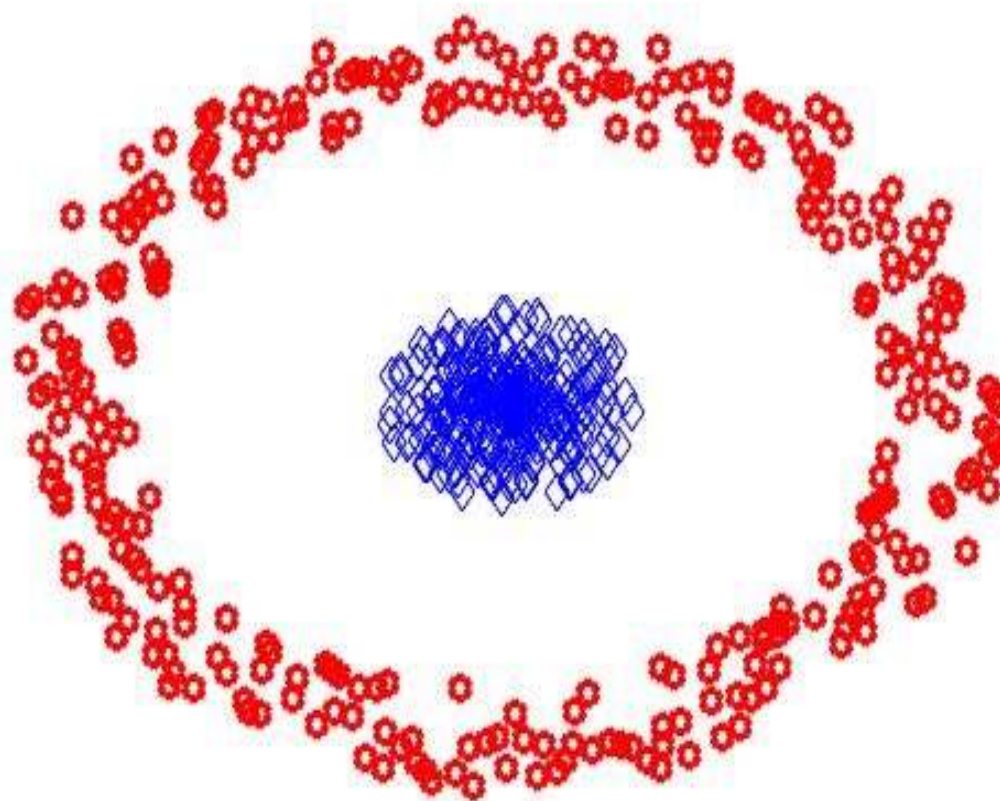
# Deep Networks

- Will the networks **generalize** with deep networks
  - DNNs are **quite data hungry** and performance improves by increasing the data.
  - Generalization problem is tackled by **providing training data from all possible conditions**.
    - Many artificial data augmentation methods have been successfully deployed
  - Providing the **state-of-art performance in several real world applications**.

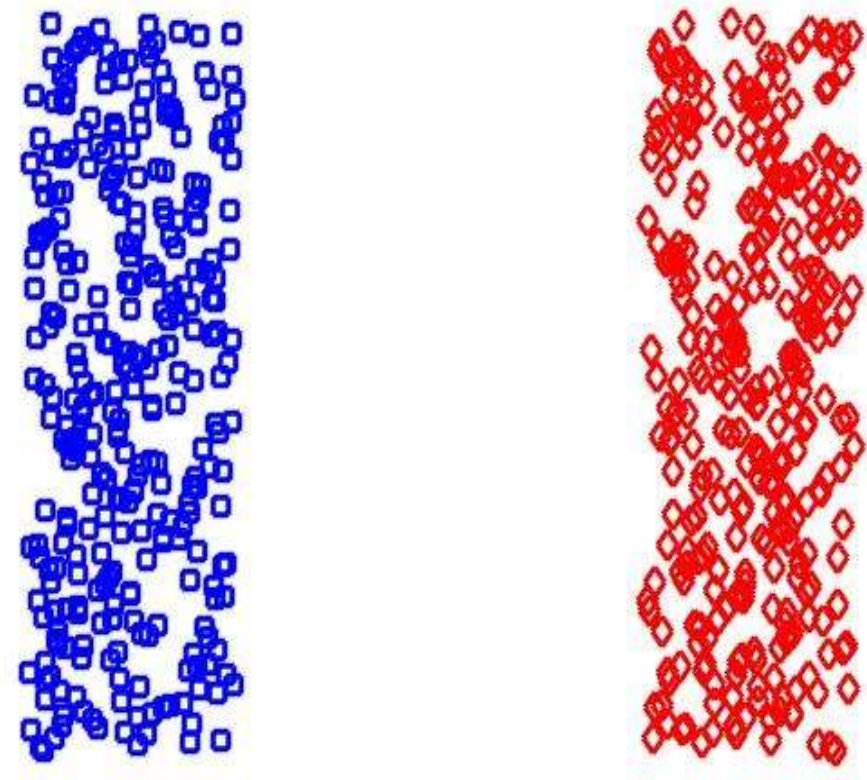
# Representation Learning in Deep Networks

- The input data representation is one of most important components of any machine learning system.

Cartesian Coordinates



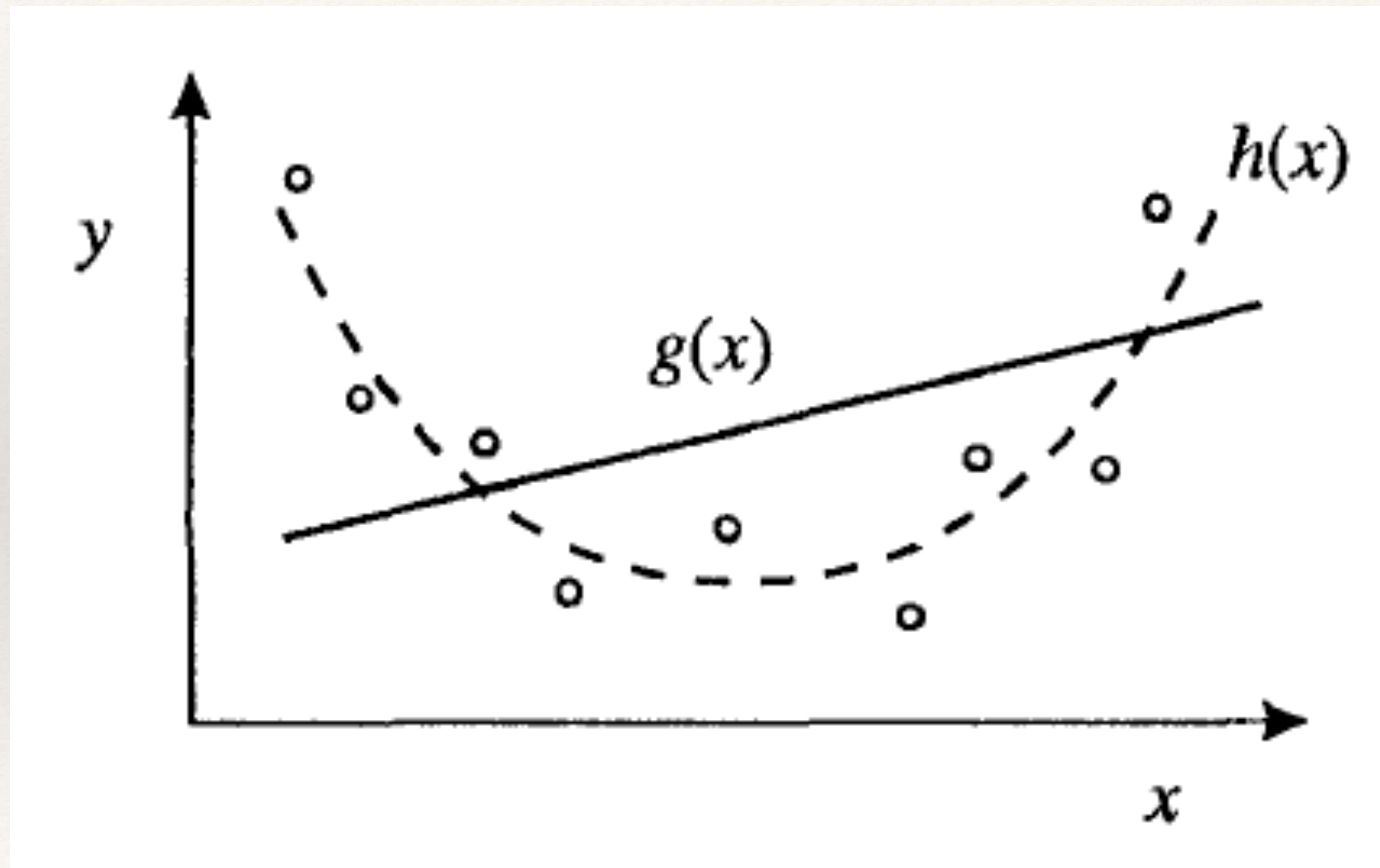
Polar Coordinates



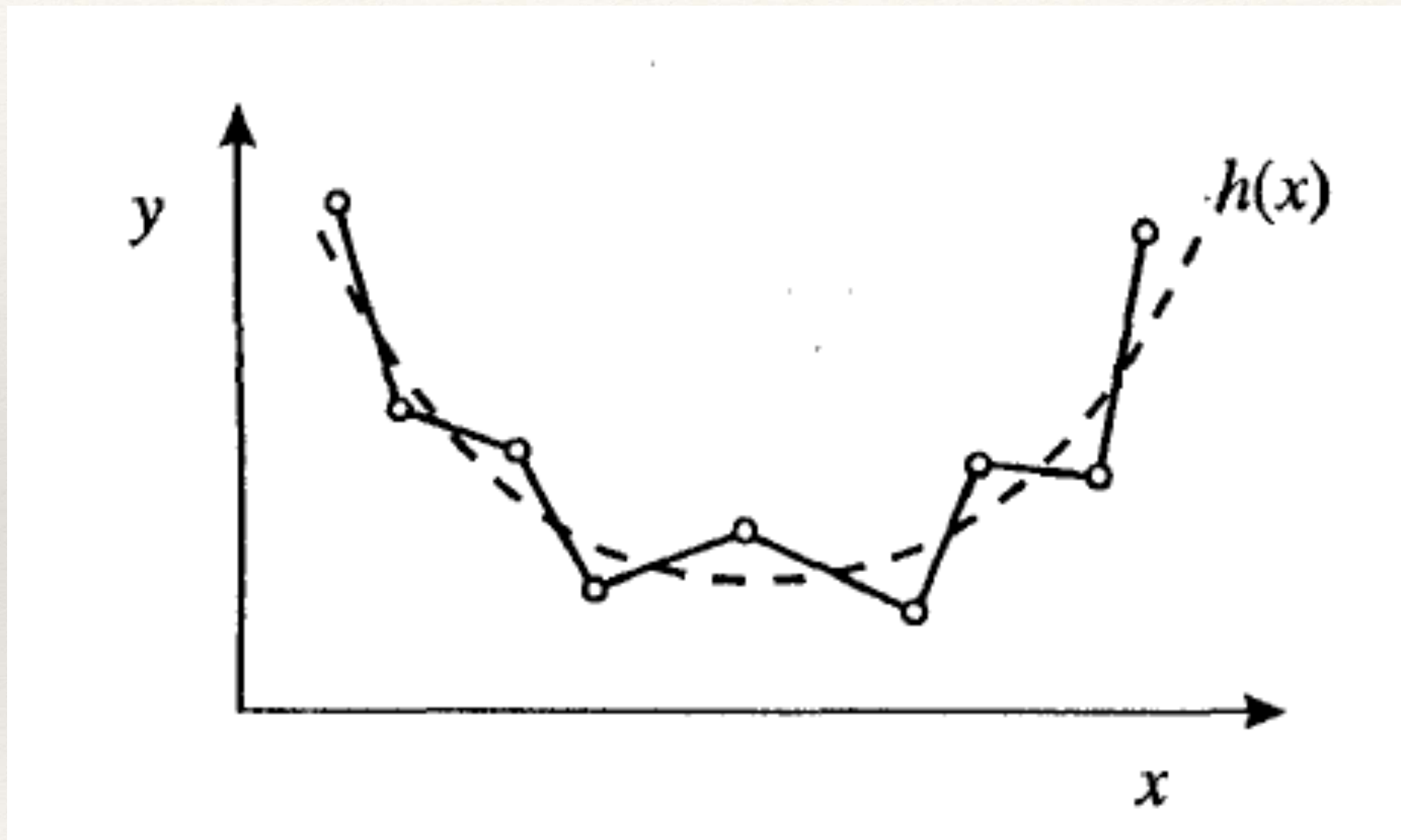
# Representation Learning in Deep Networks

- The input data representation is one of most important components of any machine learning system.
  - Extract factors that enable classification while suppressing factors which are susceptible to noise.
- Finding the right representation for real world applications - substantially challenging.
  - Deep learning solution - **build complex representations from simpler representations.**
  - The dependencies between these hierarchical representations are refined by the target.

# Underfit



# Overfit



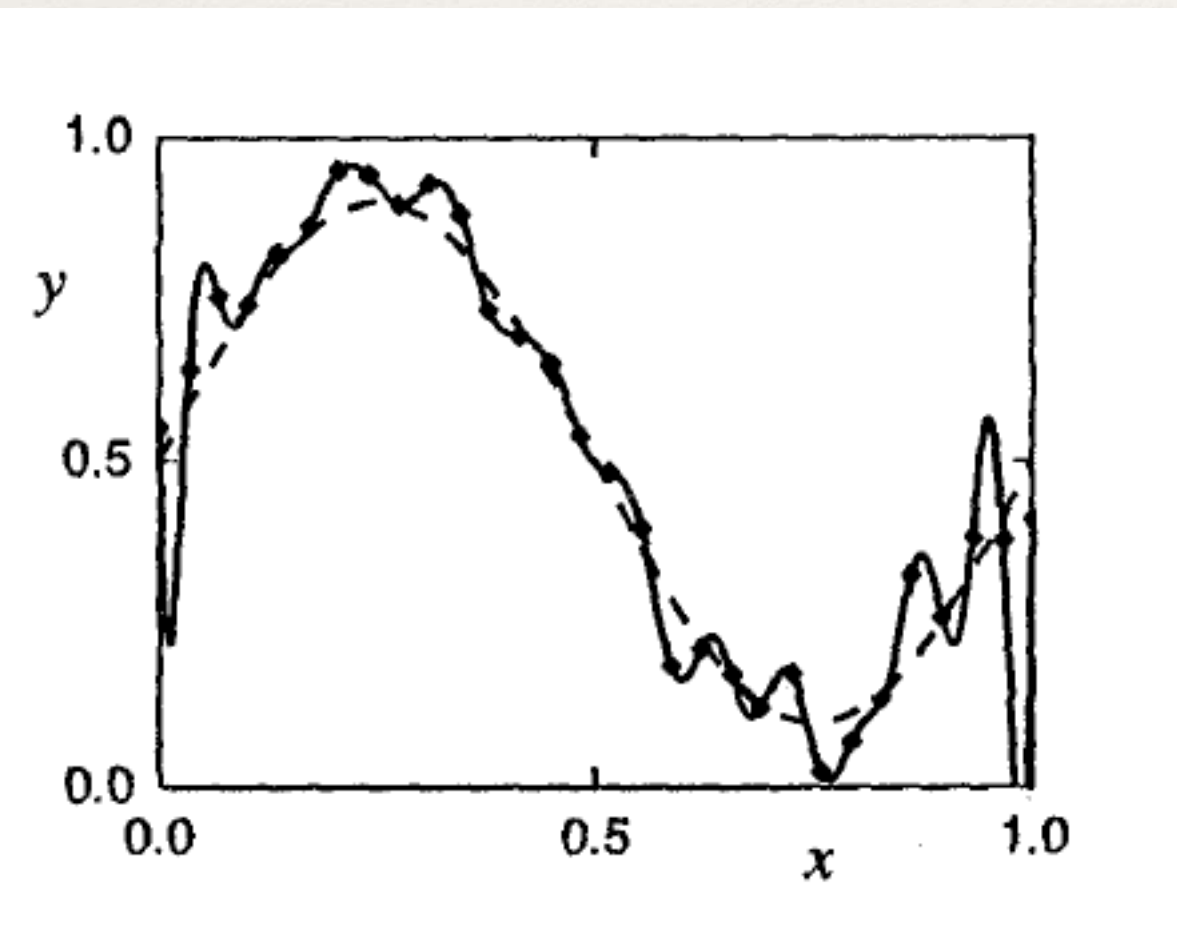
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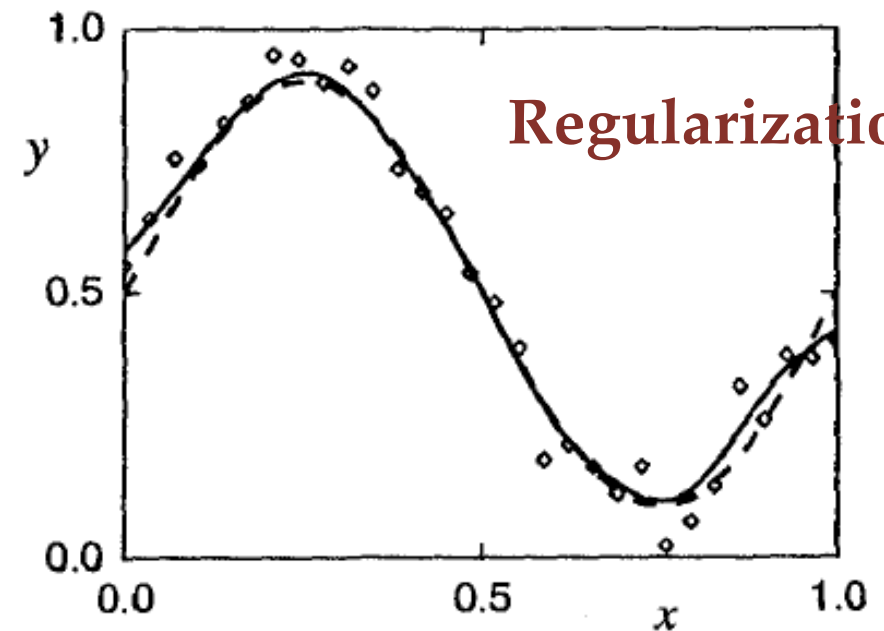
# Avoiding OverFitting In Practice

# Weight Decay Regularization

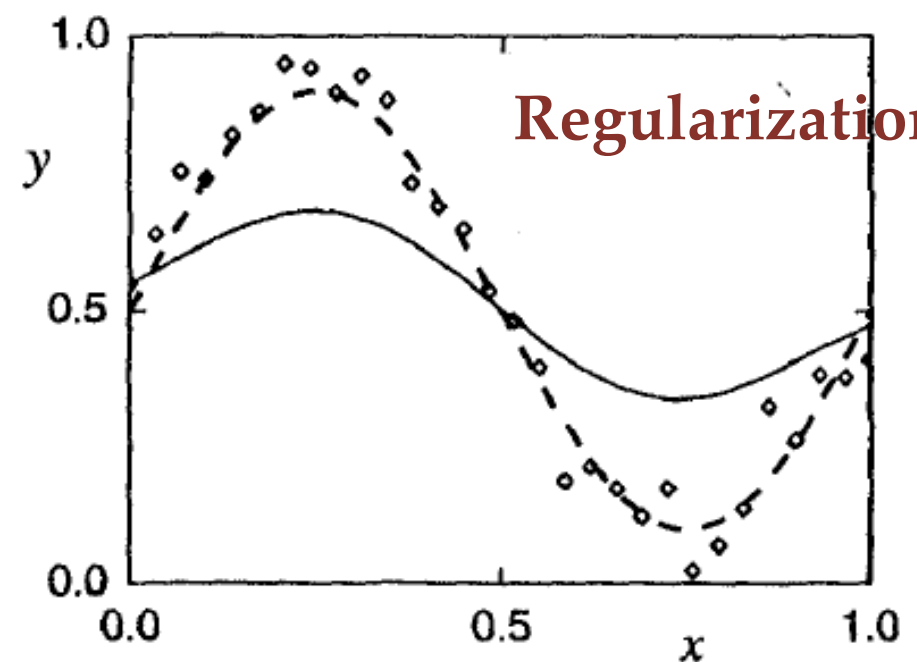
Regularization = 0



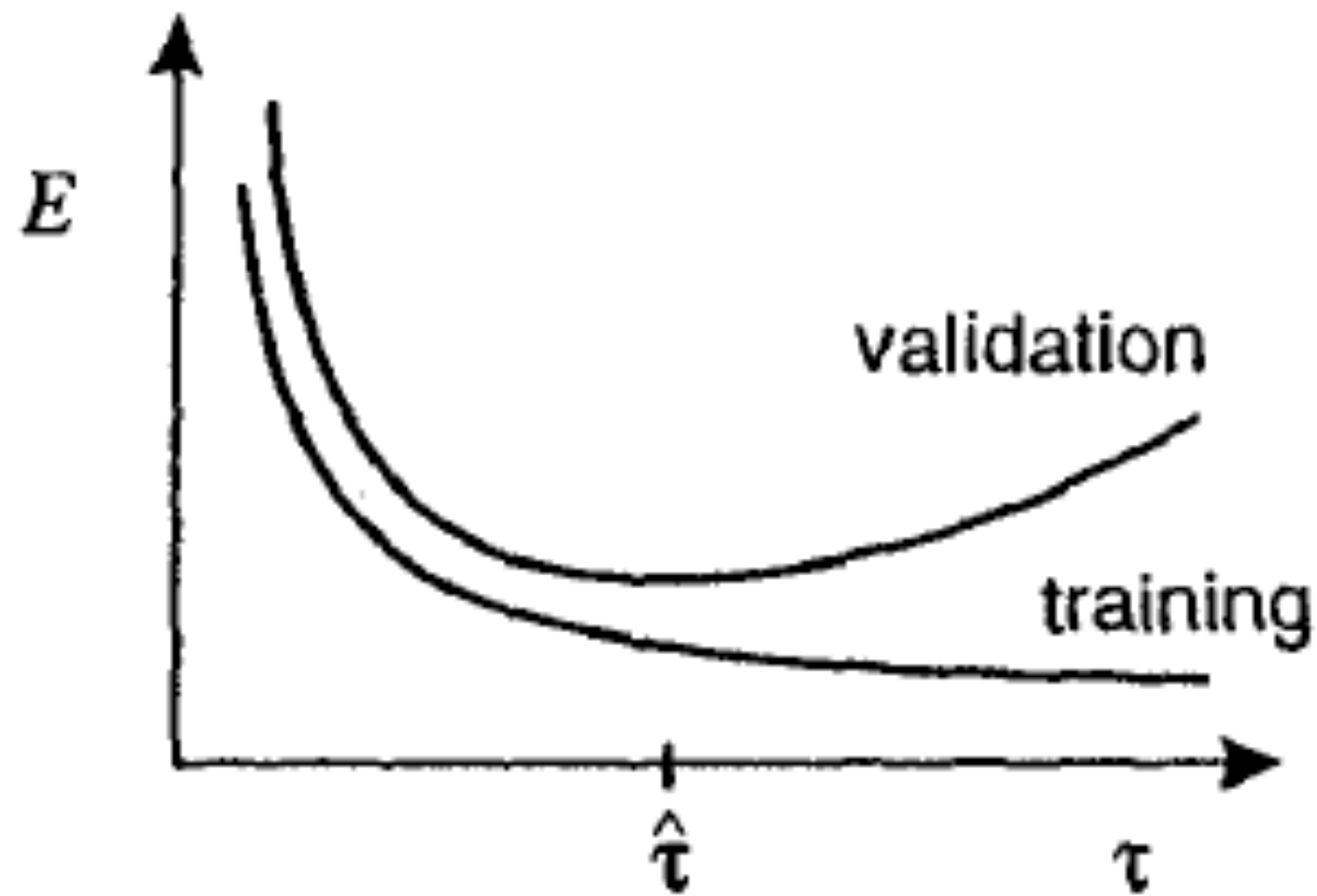
Regularization = 40



Regularization = 4000



# Early Stopping



Most Popular in Practice

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# Batch Normalization

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Batch Normalization: Accelerating Deep Network Training by  
Reducing Internal Covariate Shift

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Christian Szegedy

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# Effect of Batch Normalization

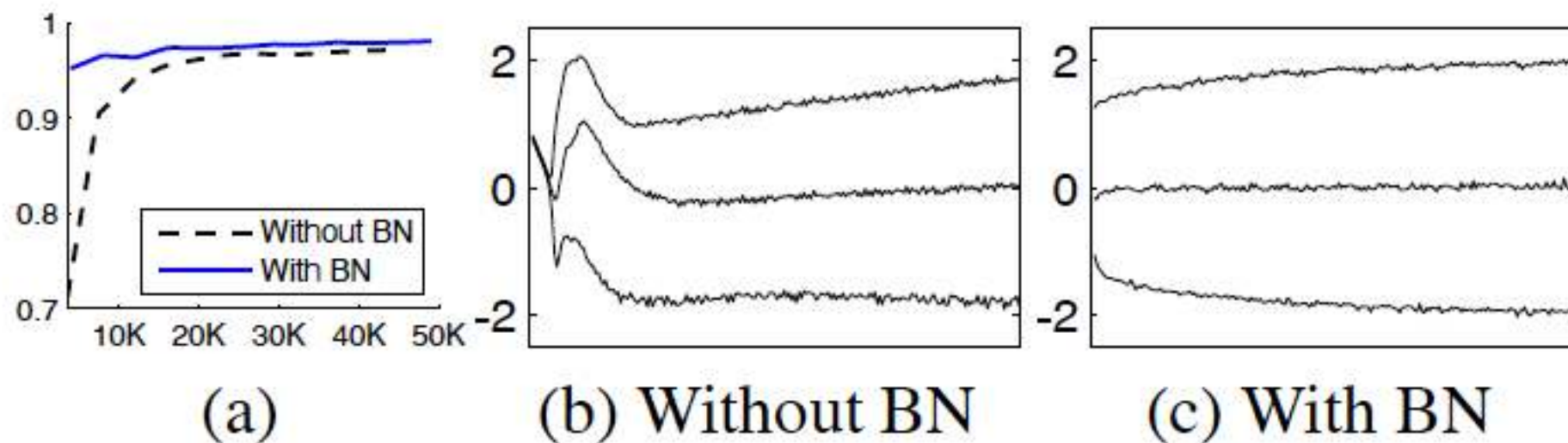


Figure 1: (a) *The test accuracy of the MNIST network trained with and without Batch Normalization, vs. the number of training steps. Batch Normalization helps the network train faster and achieve higher accuracy.* (b, c) *The evolution of input distributions to a typical sigmoid, over the course of training, shown as {15, 50, 85}th percentiles. Batch Normalization makes the distribution more stable and reduces the internal covariate shift.*

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# Dropout Strategy in Neural Network Training

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## Dropout: A Simple Way to Prevent Neural Networks from Overfitting

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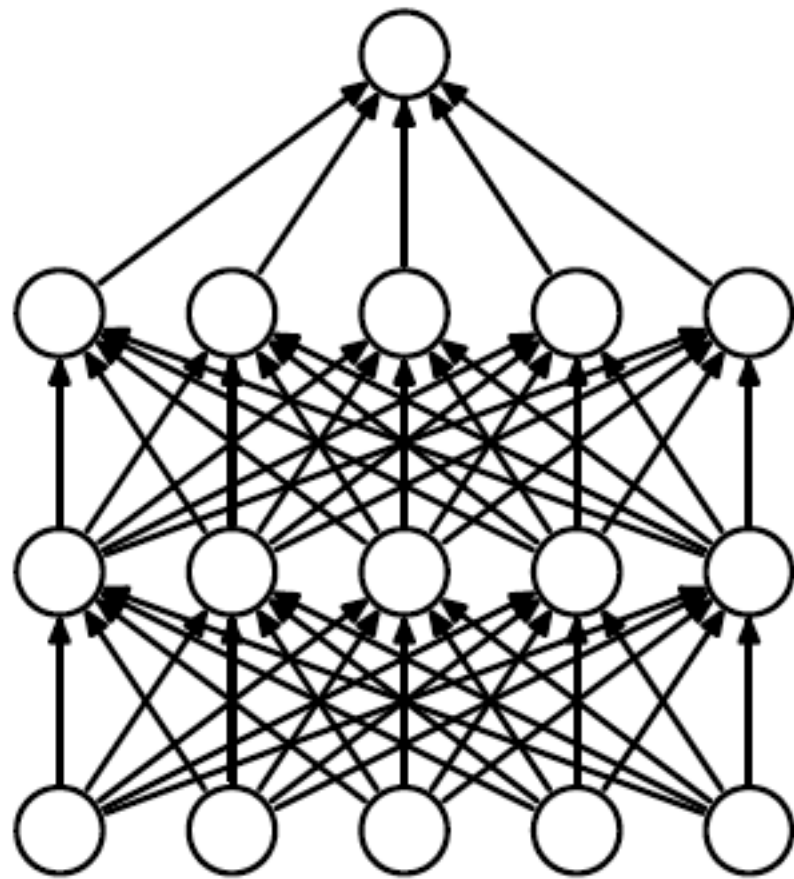
*University of Toronto*

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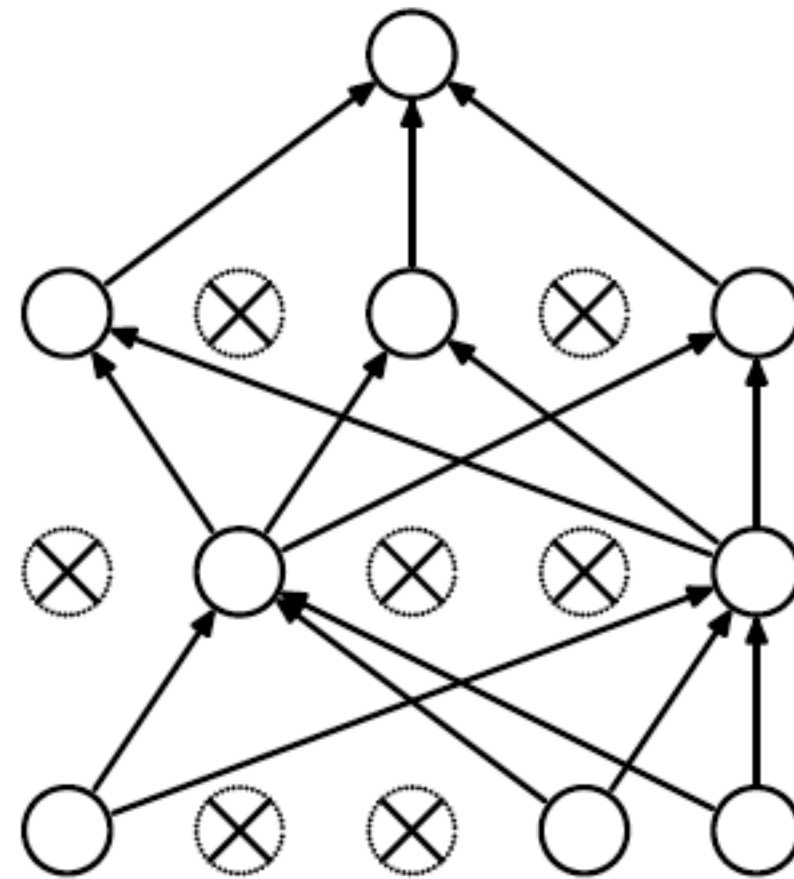
*Toronto, Ontario, M5S 3G4, Canada.*

Editor: Yoshua Bengio

# Dropouts in Neural Networks

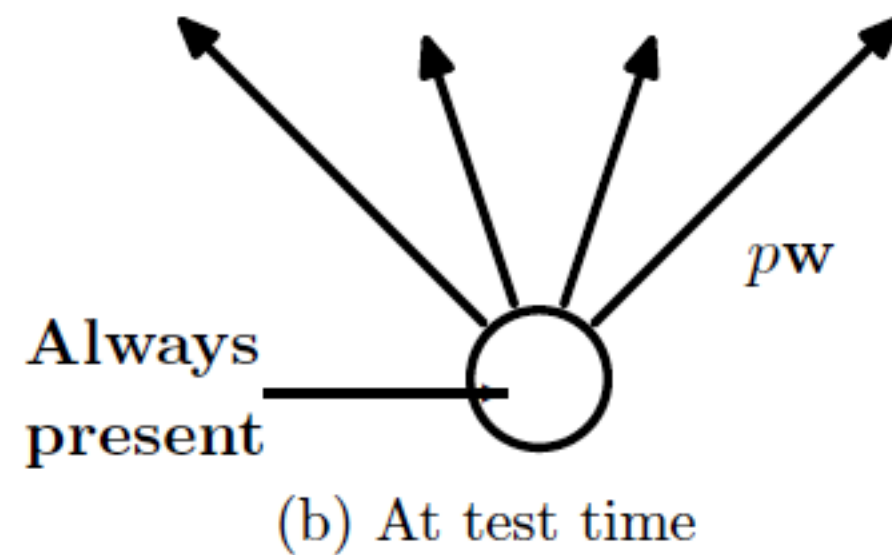
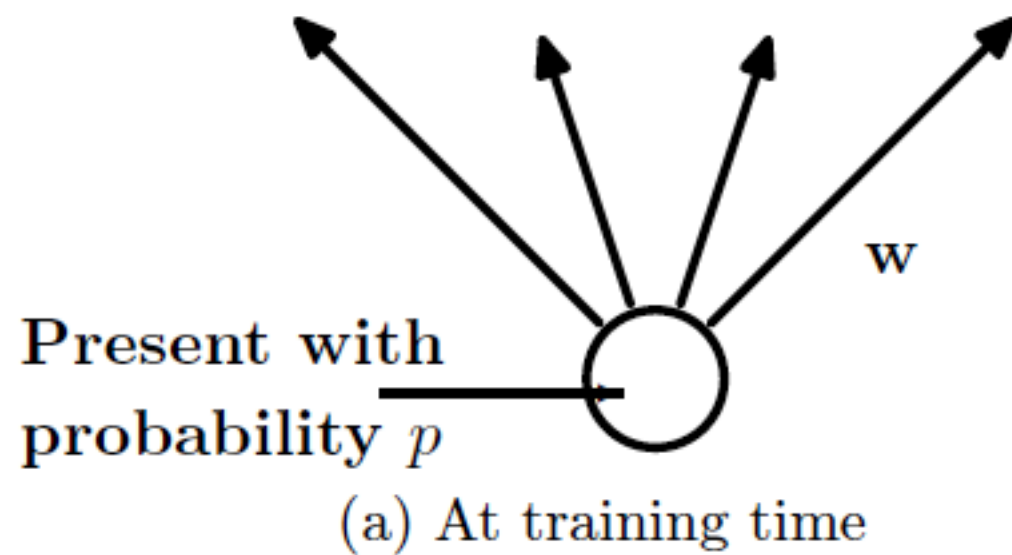


(a) Standard Neural Net



(b) After applying dropout.

# Dropout in Training and Test



# Dropout Application

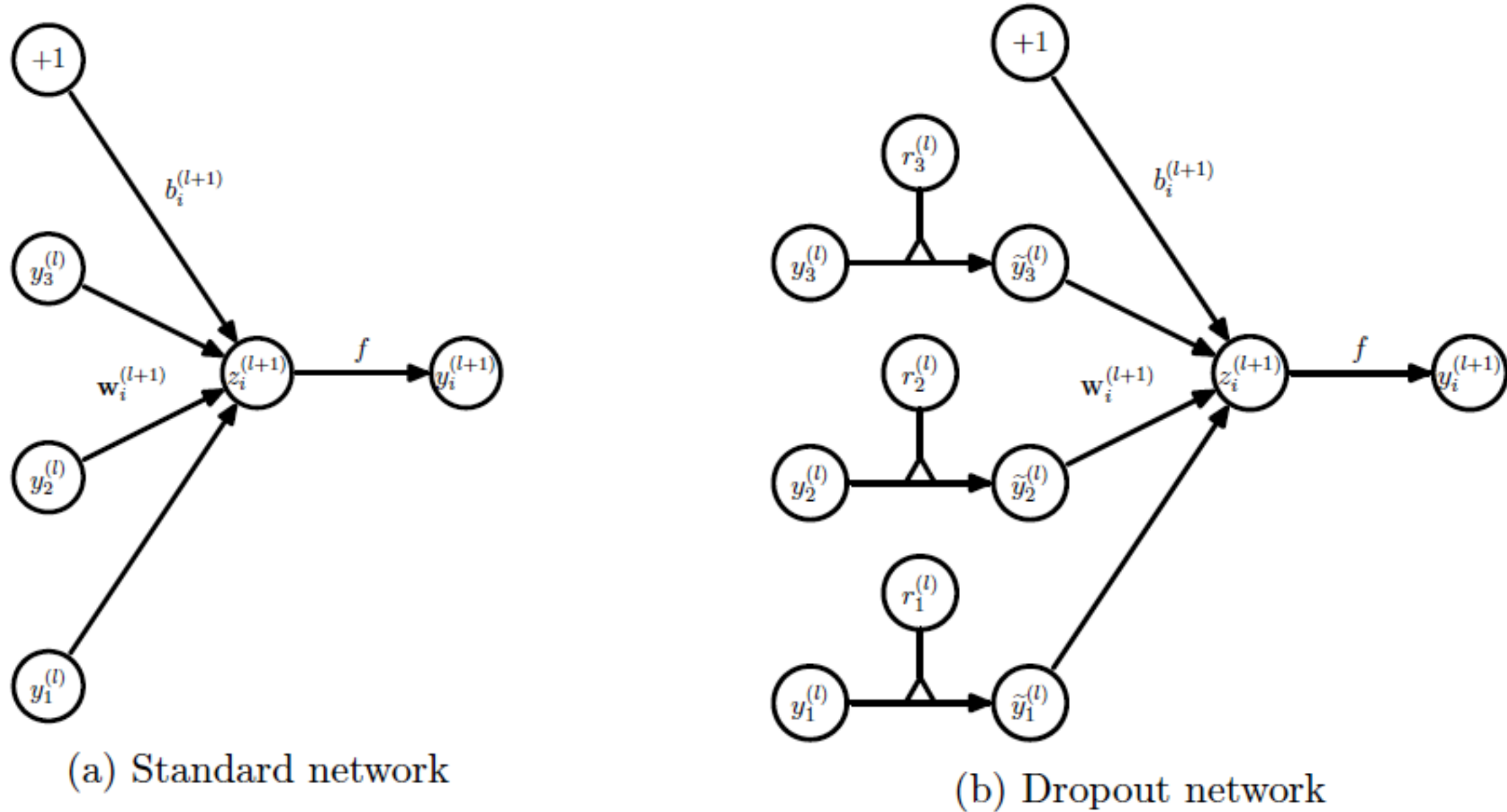


Figure 3: Comparison of the basic operations of a standard and dropout network.

# Effect of Dropouts

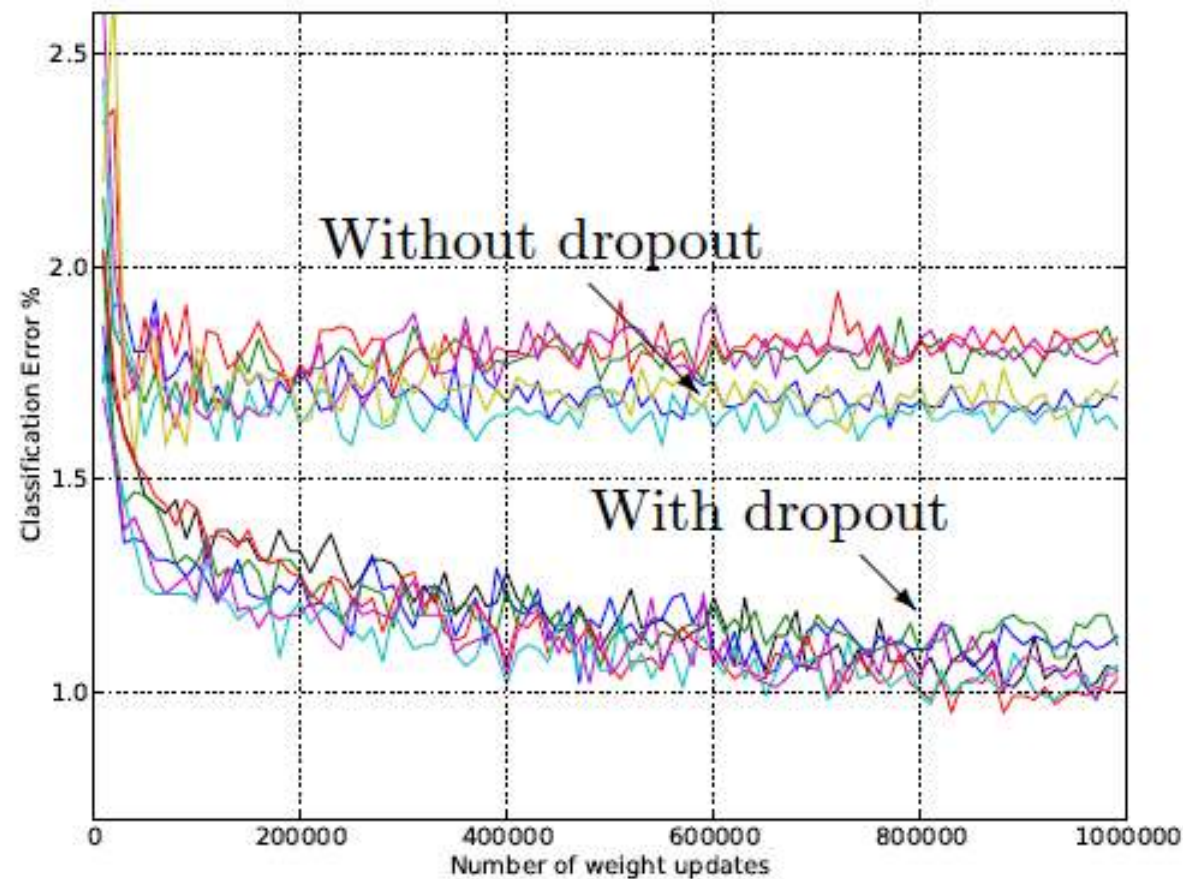


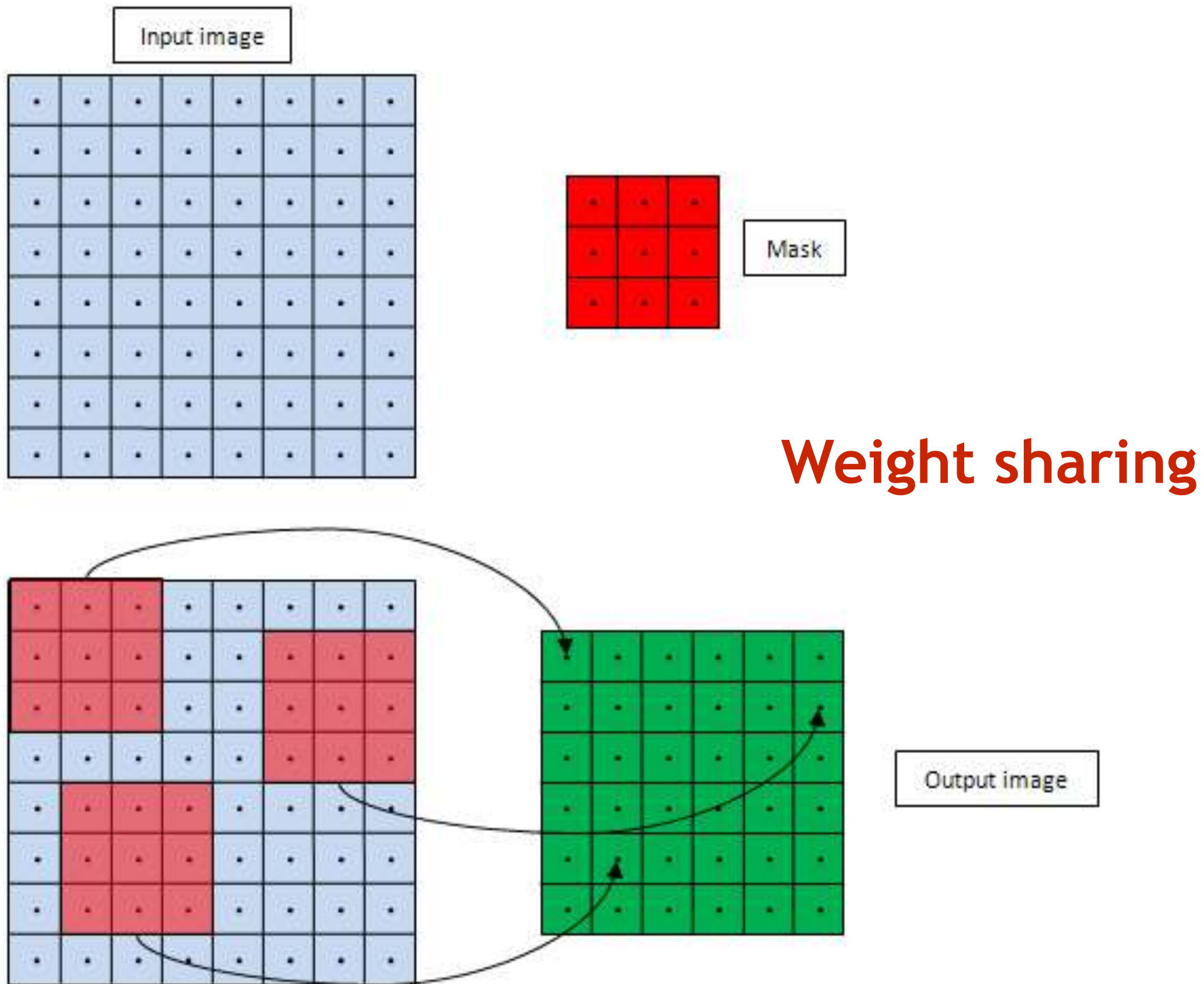
Figure 4: Test error for different architectures with and without dropout. The networks have 2 to 4 hidden layers each with 1024 to 2048 units.

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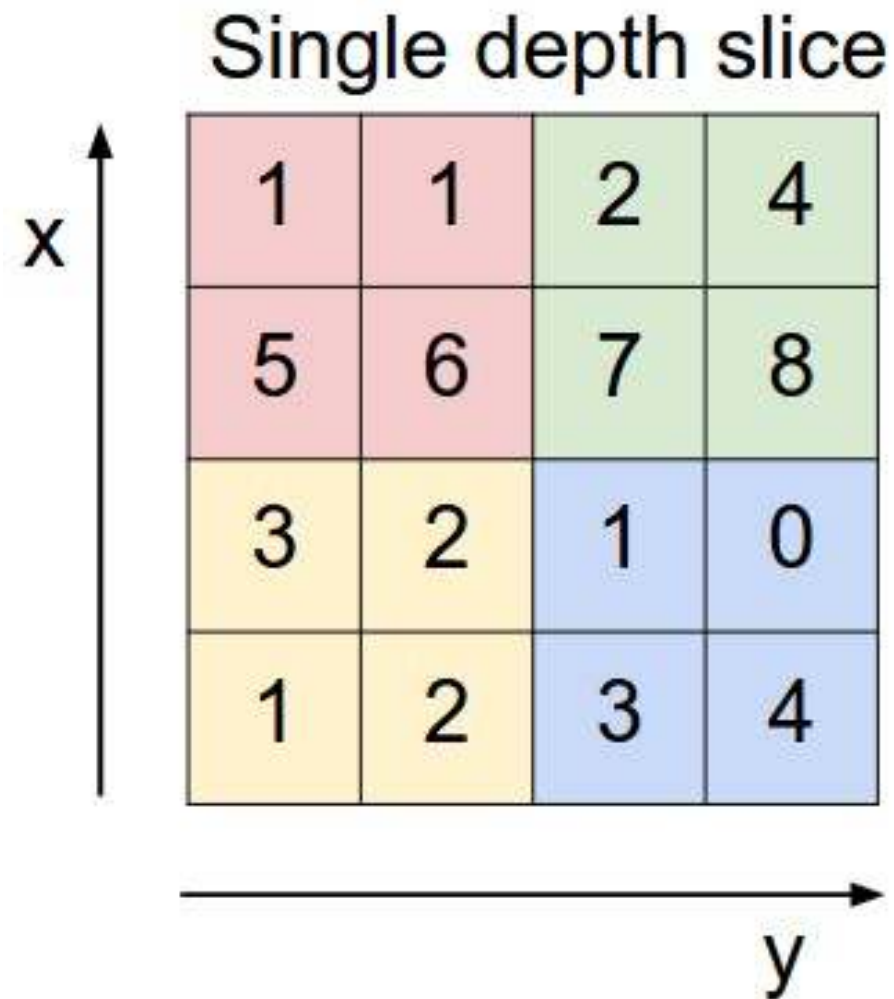
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# Convolutional Neural Networks

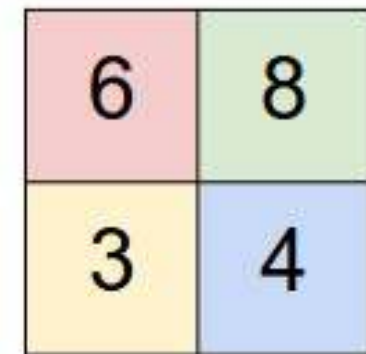
# Other Architectures - Convolution Operation



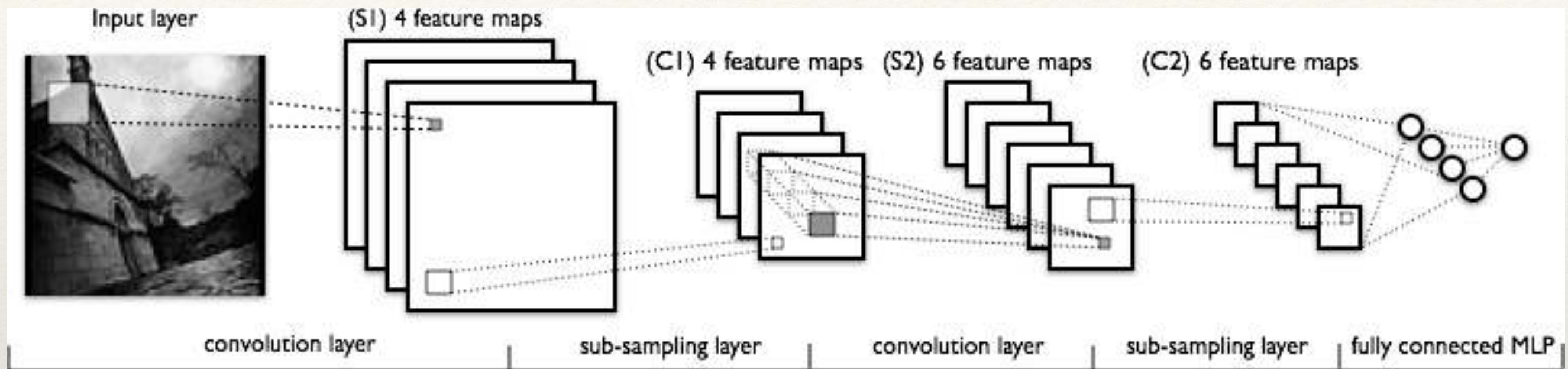
# Max Pooling Operation



max pool with 2x2 filters  
and stride 2

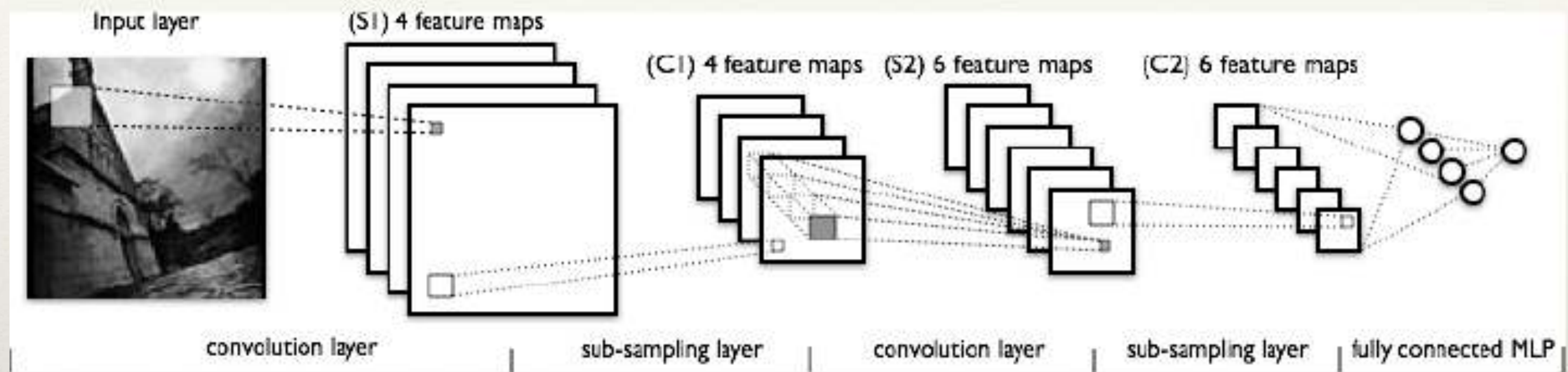


# Convolutional Neural Networks



- Multiple levels of filtering and subsampling operations.
- Feature maps are generated at every layer.

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# Back Propagation in CNNs

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